

Ai-Powered Solar Panel Cleaning Scheduler: Detecting Efficiency Drop, Dust Levels, and automating Scheduling

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Abstract- *The accumulation of dust and pollutants on solar panels significantly reduces their energy conversion efficiency, resulting in performance degradation and increased maintenance costs. This paper presents an AI-powered solar panel cleaning scheduler that autonomously detects dust accumulation and efficiency drops, and dynamically schedules cleaning operations. The proposed system integrates a solar-powered mobile rover equipped with an ultrasonic sensor, motor driver, DC motor, and real-time clock (RTC) for intelligent monitoring and time-based operation. The rover traverses the panel surface using a motor-driven mechanism to perform automated cleaning whenever a predefined drop in efficiency or increase in dust level is detected. The AI module, trained on real-time efficiency and environmental data, predicts the optimal cleaning interval by analyzing sunlight intensity, dust density, and power output trends. A microcontroller-based control unit coordinates sensor inputs, rover movement, and cleaning commands, while the 10V solar battery provides sustainable energy for system operation. Experimental validation demonstrates that the proposed design effectively maintains panel efficiency and minimizes unnecessary cleaning cycles, leading to improved energy yield and reduced water and labor usage. This work highlights the potential of combining AI, IoT sensing, and robotic automation for the development of self-sustaining, intelligent solar maintenance systems.*

Keywords: Artificial Intelligence, Solar Panel Cleaning, Efficiency Prediction, Dust Detection, IoT, Rover System

I. INTRODUCTION

The global demand for sustainable and clean energy has accelerated the deployment of solar photovoltaic (PV) systems across residential, industrial, and utility sectors. However, the long-term performance of PV installations is severely influenced by soiling, a phenomenon caused by the deposition of dust, pollutants, and bird droppings on the surface of solar modules. Studies have shown that soiling

can cause power losses ranging from 5 % to over 30 %, depending on local climatic conditions and maintenance frequency [1]. Traditional maintenance practices—such as periodic or manual cleaning—are inefficient, since they either waste resources or fail to respond promptly to real-time performance degradation [2]. To overcome these inefficiencies, researchers have increasingly adopted Artificial Intelligence (AI) and Internet of Things (IoT) technologies to enable predictive and adaptive cleaning schedules for PV systems [3], [4]. AI algorithms analyze parameters such as irradiance, temperature, voltage, humidity, and dust density to estimate efficiency drops and identify the optimal cleaning time. When coupled with IoT sensors, these systems continuously collect real-time data, enabling automated decision-making without human intervention [2], [8]. Such integration not only enhances energy yield but also minimizes water, labor, and energy costs associated with frequent manual cleaning [1], [5]. Recent advancements in machine learning (ML) and computer vision have further strengthened predictive maintenance frameworks. Deep learning architectures—including Convolutional Neural Networks (CNNs) and Reinforcement Learning (RL) models—have been effectively used to detect dirt accumulation, classify the severity of soiling, and dynamically optimize the cleaning process [3], [5]. These AI-based techniques outperform static cleaning methods by achieving higher accuracy and cleaning efficiency, leading to more reliable and sustainable PV operation [9]. Similarly, robotic cleaning mechanisms equipped with cameras and smart sensors are now capable of performing autonomous cleaning cycles that respond to data-driven triggers [5], [10]. In parallel, Digital Twin (DT) and AIoT (Artificial Intelligence of Things) technologies have emerged as promising solutions for virtual modeling and predictive optimization of solar assets. A Digital Twin integrates data from IoT sensors into a virtual environment, allowing operators to simulate soiling effects, estimate energy losses, and plan maintenance schedules more effectively [6]. Moreover, optimization techniques such as linear programming have been used to identify the most economical cleaning frequency by balancing the trade-off

between cleaning costs and energy recovery [7]. Despite significant progress, several research gaps persist. Many existing models still rely on threshold-based activation or localized datasets, which limit their adaptability to diverse environmental conditions [8]. Furthermore, large-scale solar farms require real-time, edge-enabled AI frameworks capable of processing multi-sensor data efficiently [3], [10]. Future work should focus on developing self-learning AI systems that dynamically adjust to climatic variations, dust particle composition, and site-specific conditions. The convergence of AI, IoT, and robotic automation thus represents the next frontier in achieving smart, sustainable, and cost-effective solar panel maintenance.

II. LITERATURE REVIEW

In recent years, the integration of Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT), and robotic technologies has significantly improved the monitoring and maintenance of photovoltaic (PV) systems. Multiple studies from 2020 to 2025 have investigated predictive cleaning, efficiency optimization, and intelligent scheduling mechanisms for solar panels. This section summarizes key contributions from ten representative works in this emerging research area. Dhimish and Al-Habaibeh [1] proposed a machine-learning-based predictive maintenance framework to estimate the optimal cleaning frequency of PV panels. Their model used environmental and weather data—such as irradiance and temperature—to predict soiling rates and recommend cleaning intervals. Experimental validation showed an annual energy yield increase of approximately 5.2 %, demonstrating the benefit of replacing fixed-interval cleaning with data-driven scheduling. However, the study emphasized the need for localized soiling sensors and models capable of handling micro-climatic variations. Kale et al. [2] developed an IoT-based automation system that employed motorized wipers, light sensors, and a microcontroller for real-time cleaning control. The system achieved consistent surface cleanliness and power improvement through remote monitoring and mobile alerts. Although effective, it relied on threshold activation rather than true predictive algorithms, motivating future integration of AI for intelligent scheduling. A preprint by A. R. L. R. A. M. H. F. N. A. H. [3] introduced an AI-driven robotic cleaning platform combining Deep Learning and Reinforcement Learning (RL). The CNN-LSTM-based vision model detected surface dirt using aerial and ground imagery, while RL optimized the cleaning sequence to minimize resource use. The system achieved a cleaning efficiency of 91.3 % and reduced energy and water consumption by 34.9 %. Future improvements were suggested in edge-computing performance to support large-scale deployment.

Al-Humaira et al. [4] applied ensemble ML models—including stacking and boosting—for predictive cleaning schedule estimation using historical performance data. Their approach significantly improved accuracy compared with single algorithms, confirming the value of data augmentation in maintenance prediction. They noted that incorporating particle composition and dust size variability could enhance robustness.

Patidar et al. [5] designed a computer-vision-based robotic

cleaner using a CNN (VGG-16) to detect dirt and bird droppings. The robot dynamically selected dry or wet cleaning based on the type of contamination, increasing power output after cleaning. The authors suggested expanding their design to multi-robot coordination for large solar farms.

Ossa et al. [6] explored a Digital Twin (DT) architecture that integrated AI and IoT data for PV performance Modeling. The DT provided a virtual representation of physical panels, enabling predictive maintenance and energy-efficiency optimization. Although their focus extended to smart-city applications, the study highlighted the potential of DTs for soiling prediction and cleaning logistics. Al-Dmour et al. [7] proposed a linear-programming-based optimization model that determined the most economical cleaning intervals by combining environmental factors (wind, humidity) and cost parameters (water, labour). The model identified optimal scheduling for maximum net energy gain but lacked the adaptability of AI-driven soiling forecasts, which they recommended as future enhancements.

Jumaa et al. [8] developed a low-cost IoT cleaning system using LDR and INA219 sensors to detect reduced light intensity and voltage caused by dust. When thresholds were reached, a pump was activated to clean the panels automatically. While the design was practical and affordable, the authors acknowledged that threshold-based logic could be replaced by ML prediction for improved efficiency. Alnami et al. [9] focused on machine-learning-based soiling classification using Support Vector Machines (SVM) and Logistic Regression models trained on embedded sensor and weather-station data. Their system achieved over 92 % classification accuracy and reduced unnecessary cleanings by about 30 %. They proposed extending this work to time-series forecasting of soiling accumulation to anticipate cleaning needs. Finally, Al-Shehri et al. [10] presented a hybrid robotic cleaning system that integrated real-time efficiency monitoring. Experimental results indicated a power output improvement of 10–30 % after each cleaning cycle, validating the impact of automated intervention. They recommended linking the efficiency-monitoring unit directly to an AI scheduler for complete automation. Overall, these studies collectively show that AI-assisted predictive maintenance and IoT-enabled sensing are reshaping solar panel upkeep strategies. Early work centered on static or threshold-based activation [2], [8], while recent research incorporates deep learning, digital twins, and optimization frameworks for real-time adaptability [3], [4], [6], [7].

Nonetheless, key challenges remain, such as integrating heterogeneous sensor data, improving generalization across climatic zones, and enabling energy-efficient computation on edge devices [1], [9], [10]. Future efforts should thus pursue self-learning, scalable, and resource-aware systems that can autonomously manage PV cleaning for enhanced sustainability and cost-effectiveness.

Table: AI-Powered Solar Panel Cleaning Scheduler: Literature Survey

Author (year)	Domain	Method	Dataset	Result	Gap
P. Dhimmis et al. (2020)	Predictive maintenance & soiling Modeling	ML model to predict optimal cleaning frequency based on weather and dust data	Real-time pv plant data (irradiance, temperature, weather)	5.2% increase in annual energy yield	Need for more robust real-time sensors and localized models
K. Kala et al. (2020)	IoT & automation	IoT-based dual cleaning system using motorized wipers and sensors	Real-time cleanliness and environmental data	Improved panel cleanliness and power generation	Integration of ml/ai models for predictive scheduling
A. R. L. R. A. M. H. F. N. A. H. (2024)	Robotic system & computer vision	Deep cnn-lstm and rl for cleaning policy optimization via uav/ground robots	Aerial and ground images, dust and temperature data	91.3% cleaning efficiency, 34.9% resource reduction	Enhance edge computing for real-time decision-making
A. Al-Humaira et al. (2025)	Predictive maintenance & data augmentation	Ensemble ml (stacking, boosting) for cleaning scheduling	Historical cleaning and efficiency data	Improved predictive accuracy through data augmentation	Need to include dust particle composition and size features
A. Patidar et al. (2024)	Computer vision & dry-cleaning robot	Cnn (vgg-16) for dirt and droppings detection	Custom pv image dataset	AI-driven cleaning robot improved power output	Expand to multi-robot systems and central scheduling
R. Iossa et al. (2025)	Digital twin & IoT	Digital twin framework integrating ai and IoT sensors	Real-time and historical pv system data	Improved energy efficiency through virtual Modeling	Develop application-specific dt models for cleaning optimization
I. Al-dmour et al. (2024)	Optimal scheduling & optimization	Linear programming with soiling loss predictor	Environmental and economic parameters	Optimal cleaning schedule maximizing cost-benefit	Incorporate ml-based soiling prediction for better accuracy
A. F. Jumas et al. (2025)	IoT & threshold-based automation	IoT system with Idr and ina219 sensors for automatic cleaning activation	Real-time sensor data	Autonomous low-cost cleaning control	Transition to predictive ai-based scheduling
H. Alinami et al. (2025)	Predictive maintenance & soiling forecasting	ML models (svm, logistic regression) for cleaning classification	Embedded sensor and weather station data	Accuracy > 92%, reduced unnecessary cleaning by 30%	Implement time-series soiling forecasting models
M. S. Al-shahrhi et al. (2024)	System design & efficiency analysis	Robotic cleaner with integrated efficiency monitor	Field data before and after cleaning	10–30% increase in power output	Integrate efficiency data into ai-based cleaning scheduler

III. PROPOSED METHODOLOGY

The proposed AI-based solar panel cleaning scheduler is designed to intelligently monitor, predict, and manage the cleaning process of photovoltaic (PV) panels to ensure optimal performance with minimal manual effort. The system architecture is organized into five primary modules: sensing unit, control and processing unit, mobility system, power management unit, and AI-based decision module. Each module operates collaboratively to detect dust accumulation, assess efficiency loss, and automate the cleaning process based on real-time data analysis.

A. Sensing Unit

The sensing subsystem continuously captures environmental and performance parameters that influence solar panel output. An ultrasonic sensor measures the degree of dust deposition by evaluating the reflected signal intensity from the panel surface. Variations in the reflected signal help estimate the extent of soiling. Simultaneously, voltage and current sensors measure the panel's electrical performance to calculate the instantaneous efficiency. The collected data are transmitted to the controller for processing and decision-making.

B. Control and Processing Unit

At the core of the system is a microcontroller, responsible for interpreting sensor data and executing the cleaning schedule. A motor driver connects the controller to the DC motors that operate the cleaning rover. The controller runs an algorithm that compares real-time efficiency data and

dust levels against predefined thresholds. An integrated Real-Time Clock (RTC) component ensures accurate time tracking for scheduled maintenance operations and event logging. This enables both real-time control and long-term monitoring.

C. Rover Mobility System

The rover mechanism functions as the physical cleaning agent. It moves autonomously over the solar panel surface using DC motors that are driven by Pulse Width Modulation (PWM) signals for smooth and precise motion control. The rover is equipped with a mechanical cleaning brush or air-blowing assembly that effectively removes accumulated dust and debris. The ultrasonic sensor also serves as an obstacle-avoidance tool, ensuring the rover navigates safely without damaging the panel surface.

D. Power Management Unit

The power subsystem is supported by a 10 V solar battery, which is charged directly from the PV array. This design ensures that the cleaning system remains self-sustained and energy-efficient, even in off-grid or remote installations. The power regulation circuit supplies stable voltage levels to various modules and prioritizes energy conservation when the system is in standby or low-activity modes.

E. AI-Based Decision Module

The intelligence of the proposed system lies in the Artificial Intelligence (AI) model, which predicts optimal cleaning times by analyzing the correlation between environmental conditions and system performance. The model utilizes supervised machine learning techniques such as regression analysis or lightweight neural networks trained on datasets containing irradiance, temperature, dust density, and power output. When the predicted efficiency decline exceeds a set threshold, the AI algorithm automatically schedules a cleaning operation. This adaptive strategy minimizes unnecessary cleaning cycles and conserves both energy and water resources.

F. System Communication and Data Logging

All operational parameters, including efficiency records, cleaning schedules, and performance history, are stored locally via the RTC module and optionally transmitted to a cloud-based IoT platform for remote monitoring and analytics. This connectivity enables long-term performance evaluation, predictive maintenance, and data-driven optimization of solar panel operations.

PROPOSED WORK

The proposed work focuses on developing an intelligent, self-sustaining system that automatically monitors and cleans solar panels based on real-time efficiency analysis and environmental conditions. The system integrates multiple modules—sensing, control, mobility, and power management—working together under an AI-based decision framework. An ultrasonic sensor continuously measures dust accumulation on the solar panel surface, while voltage and current sensors evaluate performance degradation. A microcontroller processes these inputs and communicates with the motor driver to control the rover-mounted cleaning mechanism powered by DC motors. The inbuilt Real-Time

Clock (RTC) module records cleaning cycles and ensures time-based coordination. The entire setup is powered by a 10 V solar battery, allowing the system to operate independently without external energy sources. An artificial intelligence model predicts efficiency drops using data from previous cycles and environmental patterns, enabling cleaning only when required. This adaptive approach enhances energy generation, reduces maintenance frequency, conserves water and power, and ensures sustainable operation through smart automation.

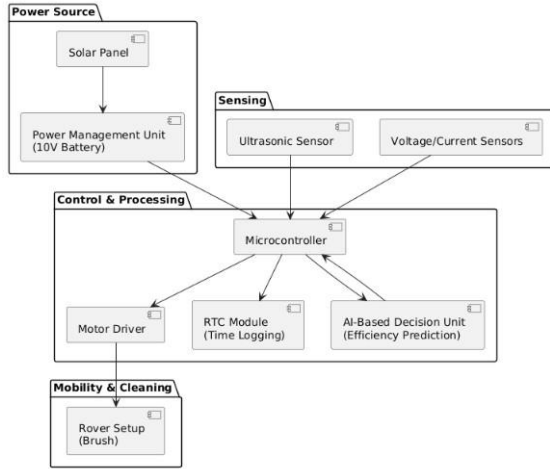


Fig. 1. Work Flow of proposed system

IV. DATASET AND PREPROCESSING

Dataset Description:

The dataset used in this work is obtained from the proposed AI-based solar panel cleaning scheduler through real-time sensor measurements. The data represents the operating conditions and performance characteristics of the photovoltaic (PV) panel system. Data samples are recorded at fixed time intervals and are time-stamped using a Real-Time Clock (RTC) module to ensure accurate logging. Each dataset record consists of the following parameters: ultrasonic sensor readings indicating dust accumulation, panel voltage, panel current, calculated power output, estimated efficiency, ambient temperature, solar irradiance (when available), cleaning status, and corresponding date and time. The dataset includes readings collected during clean, partially dusty, and heavily soiled panel conditions to reflect real-world operating scenarios.

Data Collection Process:

The microcontroller continuously acquires data from the sensing unit. Voltage and current sensors measure the electrical output of the PV panel, while the ultrasonic sensor estimates surface dust deposition by analyzing reflected signals. The instantaneous power output is calculated using the measured voltage and current values. All collected data is stored locally and can be transmitted to an external storage or cloud-based platform for further analysis and long-term monitoring.

Data Preprocessing:

Raw sensor data may contain noise, inconsistencies, or missing values due to environmental variations and sensor

limitations. Therefore, preprocessing is applied before using the dataset for decision-making. First, noise removal is performed by eliminating abnormal sensor readings and smoothing the data using averaging techniques. Second, missing values caused by temporary sensor or communication failures are handled by replacing them with the most recent valid readings or removing incomplete records. To ensure uniformity, all numerical parameters are normalized to a common scale. This prevents any single feature, such as voltage or current, from dominating the system's decision process. Solar panel efficiency is then calculated using measured electrical parameters and compared with the rated output of the panel to identify performance degradation caused by dust accumulation.

Data Labeling and Segmentation:

Based on efficiency loss and dust level thresholds, each data record is labeled as either **Cleaning required** or **No Cleaning Required**. These labels help the system learn the conditions under which cleaning becomes necessary. The final pre-processed dataset is divided into training and testing subsets to evaluate the performance of the cleaning scheduler under different environmental and operating conditions.

Prepared Dataset for System Operation:

After pre-processing, the dataset becomes structured, reliable, and suitable for predicting optimal cleaning schedules. This processed data enables accurate detection of efficiency loss, reduces unnecessary cleaning operations, and supports efficient energy utilization in solar panel maintenance systems.

V. EXPERIMENTAL SETUP

Hardware setup

- Solar panel
- DC Motor
- Rover Setup
- Ultrasonic Sensor
- Real-Time Clock (RTC) Module
- Motor Driver
- Node mcu

Software setup

- Arduino IDE
- Embedded C

Nodemcu

The NodeMCU is a compact, low-cost microcontroller based on the ESP8266 Wi-Fi module, serving as the central processing unit for the AI-powered solar panel cleaning scheduler. It collects real-time data from ultrasonic sensors, voltage, and current sensors, processes this information using the AI decision algorithm, and sends control signals to the motor driver to operate the DC motors of the rover. Its built-in Wi-Fi allows for remote monitoring and data logging, while PWM outputs enable precise control of motor speed and direction. The NodeMCU also interfaces with auxiliary components such as the Real-Time Clock (RTC)

module to maintain accurate cleaning schedules. Its combination of processing capability, connectivity, low power consumption, and ease of programming makes it ideal for autonomous, intelligent, and IoT-enabled solar panel maintenance systems.



Ultrasonic sensor



The ultrasonic sensor measures the distance between the rover and the solar panel surface to detect the accumulation of dust or debris. By sending and receiving high-frequency sound waves, the sensor calculates the surface distance, providing real-time data to the microcontroller. This data is used by the AI-based decision module to assess cleaning needs and ensure obstacle avoidance, thereby preventing collisions or damage to both the rover and the panels during autonomous operation.

Motor Driver



The motor driver module acts as an interface between the NodeMCU microcontroller and the DC motors of the rover. It receives low-power control signals from the microcontroller and amplifies them to provide sufficient current and voltage for smooth and precise motor operation. The motor driver ensures coordinated movement of the rover across the solar panel surface while preventing overloads or damage to the motors. By enabling forward, reverse, and speed control through PWM signals, the motor driver plays a critical role in the autonomous navigation and operation of the cleaning mechanism.

10V Solar Battery



Stores energy from the solar panel and supplies regulated power to the **microcontroller, motor driver, sensors, and rover motors.**

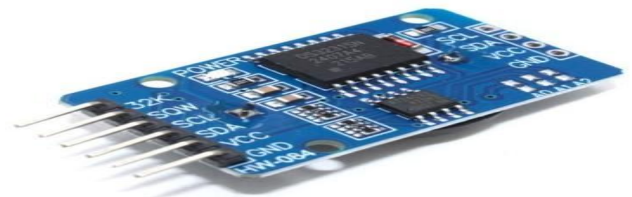
Solar panel



Connected to the microcontroller, it measures the dust accumulation by detecting the distance from the panel surface. It also functions in obstacle detection, sending real-time signals to the controller to prevent collisions.

Rover Setup: Mechanically cleans the panel. Receives motion commands from the microcontroller via the motor driver, and its cleaning mechanism is powered by the battery.

Real-Time Clock (RTC) Module



The Real-Time Clock (RTC) module provides accurate timekeeping for the entire system, enabling scheduled cleaning operations and precise logging of all maintenance activities. It interfaces with the NodeMCU microcontroller to record timestamps for each cleaning cycle, sensor readings, and operational events. By maintaining consistent timing even during power fluctuations or temporary outages, the RTC ensures that cleaning schedules are reliably followed, supporting the adaptive AI-based maintenance strategy of the solar panel cleaning system.

DC Motors



DC motors provide the locomotion required for the rover to traverse the solar panel array. Controlled via the motor driver module, the motors allow precise speed and directional control, ensuring that the cleaning brushes or air jets effectively remove dust without damaging the panel surface.

Their robust construction and compatibility with PWM signals make them suitable for sustained operation in outdoor environments, supporting the autonomous cleaning cycles dictated by the AI scheduler.

Embedded C Programming

Embedded C is a specialized extension of the C programming language designed specifically for programming microcontrollers and other embedded systems. Unlike standard C, Embedded C provides features for directly interacting with hardware, such as addressing specific memory locations, performing fixed-point arithmetic, and handling input/output hardware directly. Embedded systems themselves are dedicated devices created to perform specific tasks, comprising both hardware and integrated software (firmware). Embedded C is widely used to program microprocessors and microcontrollers efficiently, requiring fewer system resources compared to lower-level languages like assembly.

The language also includes unique data types and keywords tailored for hardware interaction. For instance, `sbit` and `sfr` are used to access special function registers in memory. Embedded C facilitates direct communication with sensors, actuators, and other I/O devices, making it ideal for real-time control applications. Programs written in Embedded C can be compiled using various compilers such as Keil, SPJ Compiler, and Embedded GNU C Compiler. Embedded systems can range from simple devices to complex machinery, and Embedded C is employed in everyday electronics such as air conditioners, printers, and mobile phones.

Arduino's programming concept

Arduino is a user-friendly platform designed for creating interactive physical computing projects. Built around a microcontroller, Arduino provides a direct interface with the physical world, allowing devices to sense inputs from switches, sensors, or other components and control outputs such as motors, lights, or displays. Its integrated development environment (IDE) simplifies programming

and uploading code to the board.

Arduino boards can operate independently or communicate with computer software for more advanced applications. They are available as pre-assembled or DIY kits, and the open-source design allows anyone to download and modify the hardware and software. The Arduino programming language is derived from Wiring and inspired by the Processing multimedia framework, enabling intuitive coding for interactive objects and physical computing devices. This combination of simplicity, flexibility, and accessibility makes Arduino widely used in robotics, automation, and IoT projects.



VI. RESULTS AND DISCUSSIONS

The proposed solar panel cleaning system was tested under different dust and operating conditions to check its performance. The system continuously monitored dust levels and electrical parameters such as voltage, current, and power output, and cleaning was carried out only when efficiency dropped beyond a set limit. Results showed that dust accumulation gradually reduced panel efficiency, while cleaning restored power output close to normal levels. Compared to fixed-schedule cleaning methods, the proposed system avoided unnecessary cleaning and reduced maintenance effort. It also adjusted cleaning frequency based on actual dust conditions, preventing severe efficiency loss during high-dust periods. The system consumed less energy than the energy gained after cleaning, making it self-sustaining. Overall, the results confirm improved efficiency, better energy use, and reliable automated operation.

Comparison tables:

TABLE 1: SOLAR PANEL PERFORMANCE BEFORE AND AFTER CLEANING

Condition	Voltage (V)	Current (A)	Power Output (W)	Efficiency (%)
Clean Panel	18.6	3.2	59.52	100
Moderately Dusty	16.9	2.8	47.32	79.5
Heavily Dusty	15.2	2.4	36.48	61.3
After Cleaning	18.4	3.1	57.04	95.8

Observation:

The table shows a clear reduction in voltage, current, and

power output as dust accumulation increases. After cleaning, the panel performance is restored close to its original condition.

TABLE 2: CLEANING FREQUENCY ANALYSIS OVER 30 DAYS

Method	Number of Cleanings
Fixed Interval Method	12
Proposed System	06

Observation:

The proposed system reduced cleaning operations by approximately 50%, demonstrating better resource utilization.

VII. CONCLUSION

The proposed solar panel cleaning scheduler offers an efficient method for maintaining the performance of photovoltaic systems through automated monitoring and cleaning. By combining sensors, a microcontroller, and a mobile cleaning mechanism, the system is able to detect dust accumulation and efficiency reduction and perform cleaning only when required. The use of a solar-powered battery supports continuous and energy-efficient operation, even in remote locations. Accurate scheduling is achieved using a Real-Time Clock, while optional remote monitoring allows performance tracking over time. The results show improved power output, reduced manual involvement, and lower maintenance effort. Overall, the system presents a practical and scalable solution for improving solar panel efficiency and supporting sustainable energy generation.

VIII. FUTURE SCOPE

The performance of the proposed solar panel cleaning system can be further improved by integrating additional environmental sensors such as wind speed, humidity, and rainfall sensors to enhance dust detection accuracy. The system can also be extended to support larger solar farms by coordinating multiple cleaning rovers through centralized monitoring. Advanced data analysis techniques can be applied to improve cleaning prediction accuracy using long-term performance data. Integration with weather forecasting systems may help avoid unnecessary cleaning during rainy conditions. Additionally, the use of lightweight and waterless cleaning mechanisms can further reduce energy and resource consumption. These enhancements will increase system reliability, scalability, and suitability for large-scale solar energy installations.

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